

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Configure Security Features in Windows and Linux

Prepared by:
Kudzaishe Majeza
Junior Network Engineer – VxLEARN Networks

Mentor:
Titus Majeza
Senior Network Engineer

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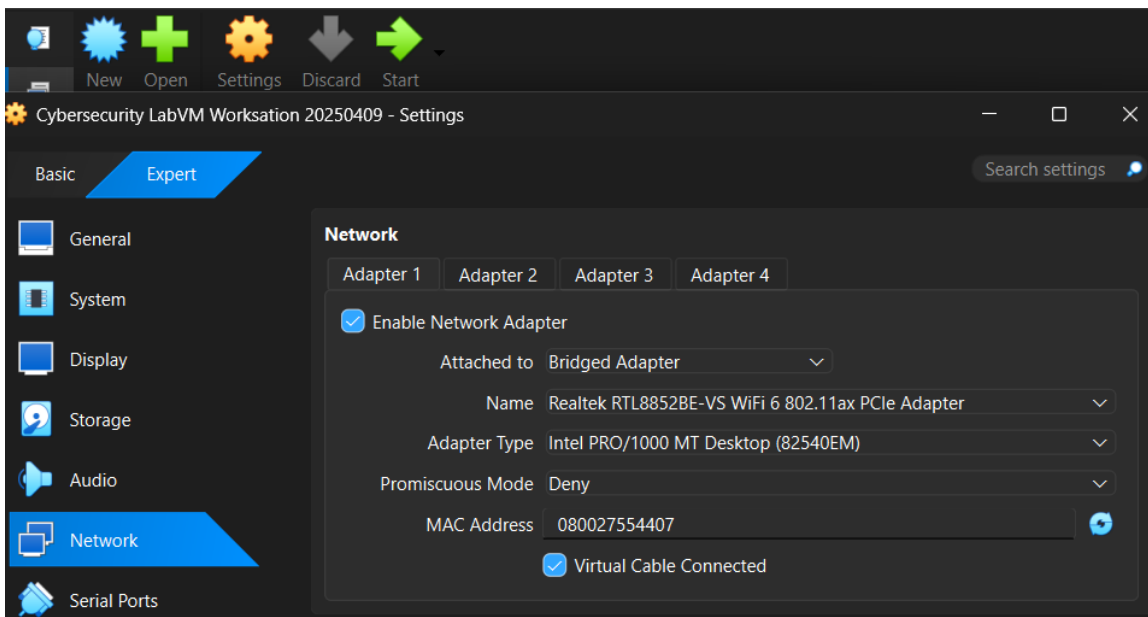
A. Introduction

This lab focused on improving system security on both Windows and Linux machines. The main goal was to update systems, configure security policies, manage firewall rules, and install security tools. These steps help protect computers from vulnerabilities and cyber threats.

B. Part 1: Updating Systems

In this section, connectivity between the Windows host and Linux virtual machine was verified using IP addresses and ping commands. The Windows system was checked for updates through the Windows Update feature, which ensures the system stays protected against known vulnerabilities. On Linux, the commands 'sudo apt-get update' and 'sudo apt-get upgrade' were used. The update command refreshed the package list, while the upgrade command installed the latest versions of software packages.

- a. Before launching the **CSE-LABVM**, select it and then choose **Settings > Network**. For **Adapter 1**, change the **Attached to:** option to **Bridged Adapter**. You can then choose the adapter. Many computers have two adapters: one for wireless networks and the other for wired networks. Choose the one your computer is using to connect to the internet.



- b. Launch the CSE-LABVM and wait for it to boot.
- c. On the CSE-LABVM, open a terminal and enter **ip address** to determine its IP address.

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:44:07 brd ff:ff:ff:ff:ff:ff
    inet 192.168.8.95/24 metric 100 brd 192.168.8.255 scope global dynamic enp0s3
        valid_lft 64556sec preferred_lft 64556sec
    inet6 fd02:1e24:c5ef:9100:a00:27ff:fe55:4407/64 scope global dynamic mngtmpa
        valid_lft 7161sec preferred_lft 3561sec
    inet6 fe80::a00:27ff:fe55:4407/64 scope link
        valid_lft forever preferred_lft forever
[analyst@secOps ~]$
```

- d. On your Windows host computer, open a command prompt and enter **ipconfig** to determine its IP address.

```
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kmaje>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : fd17:625c:f037:2:7e3f:e53c:e31a:9448
    Temporary IPv6 Address. . . . . : fd17:625c:f037:2:e05c:54a0:57c8:5a3c
    Link-local IPv6 Address . . . . . : fe80::6e6e:6b66:c142:d18e%6
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%6
                                10.0.2.2

C:\Users\kmaje>
```

Record the IP address for both CSE-LABVM and Windows PC.

CSE-LABVM: 127.0.0.1

Windows PC: 10.0.2.15

- e. From the respective command prompts, ping a website of your choice to verify that the Windows host and **CSE-LABVM** can reach the internet.

```
[analyst@secOps ~]$ ping google.com
PING google.com (192.178.54.14) 56(84) bytes of data:
64 bytes from tzjnba-ac-in-f14.1e100.net (192.178.54.14): icmp_seq=1 ttl=117 time=26.1 ms
64 bytes from tzjnba-ac-in-f14.1e100.net (192.178.54.14): icmp_seq=2 ttl=117 time=43.3 ms
```

- f. Verify that the Windows host can ping the **CSE-LABVM**.

```
C:\Users\kmaje>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time=1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

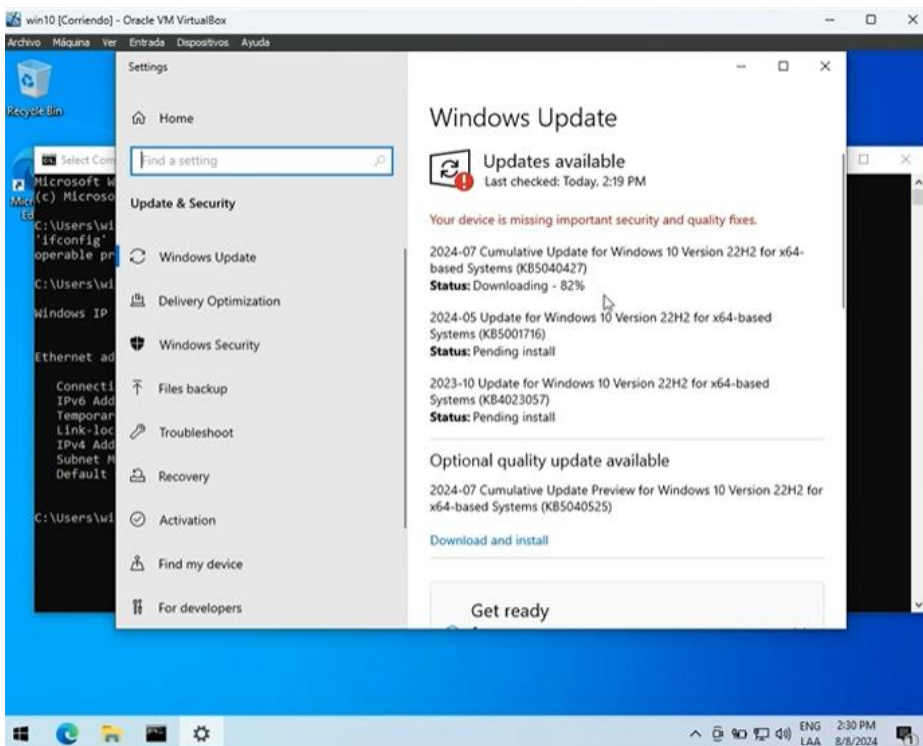
C:\Users\kmaje>
```

- g. From **CSE-LABVM**, attempt to ping the Windows host. The **CSE-LABVM** may not be able to ping the Windows host because of the default firewall settings in Windows. You will modify the firewall rule later in this lab to allow the pings through the Windows Firewall. Press **CTRL-C** to stop the pings if necessary.

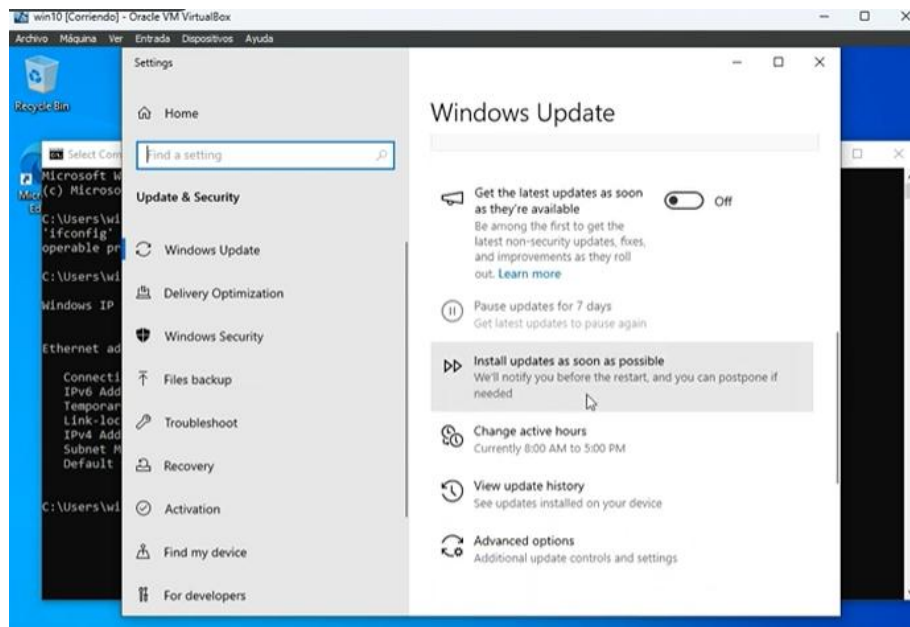
```
[analyst@secOps ~]$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
```

Step 2: Check for Windows updates.

- a. In the **Windows Start Menu**, search for **check for updates**.



- b. In the **Windows Update window**, you can review optional updates and the update history. Explore all the available options related to Windows Update and answer the following questions.



Step 3: Linux update and upgrade

- In the **CSE-LABVM**, enter the command **apt-get** to see list of available commands. The **apt-get update** command should always be done before an upgrade.
- Enter the command **sudo apt-get update** to resynchronize the package index files from their sources. Enter **password** as the password when prompted.

```
cisco@labvm:~$ sudo apt-get update
[sudo] password for cisco:
```

- In the terminal, enter the command **sudo apt-get upgrade** to retrieve and upgrade the currently installed packages with new versions available. This command will not remove currently installed packages. If the newer version cannot be upgraded, no changes will be made to the packages.

Enter **password** as the **password** if prompted. Reply **y** when asked if you want to continue. This upgrade process can take a few minutes.

```
cisco@labvm:~$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libwmf0.2-7
Use 'sudo apt autoremove' to remove it.
```

```
libreoffice-core libreoffice-style-colibre libreoffice-writer librsvg2-2 librsvg2-common
libsasl2-2 libsasl2-modules libsasl2-modules-db libsmbclient libsndfile1
libsqlite3-0 libssh-4 libssh-gcrypt-4 libssl3 libstdc++-11-dev libstdc++6 libsystend0 libtiff5
libtinfo6 libtsan0 libtsan2 libtss2-esys-3.0.2-0 libtss2-mu0 libtss2-sys1 libtss2-tcti-cmd0
libtss2-tcti-device0 libtss2-tcti-mssim0 libtss2-tcti-swtpm0 libubsan1 libunbound8
libuno-cppu3 libuno-cppuhelpergcc3-3 libuno-purpenvhelpergcc3-3 libuno-sal3
libuno-salhelpergcc3-3 libunwind8 libuuid1 libuv1 libvpx7 libvte-2.91-0 libvte-2.91-common
libwebclient0 libwebkit2gtk-4.0-37 libwebp7 libwebpdemux2 libwebpmux3 libwireshark-data
libx11-6 libx11-data libx11-xcb1 libxml2 libxpm4 libxtables12 libyajl2 linux-firmware
linux-libc-dev locales login mount ncurses-base ncurses-bin ncurses-term netplan.io
nginx-common nginx-core nmap nmap-common open-vm-tools open-vm-tools-desktop openjdk-11-jre
openjdk-11-jre-headless openssl-client openssl-server openssl-sftp-server openssl
openvswitch-common openvswitch-switch passwd perl perl-base perl-modules-5.34 procs
pulseaudio pulseaudio-utils python-apt-common python3-apport python3-apt python3-cryptography
python3-debian python3-distro-info python3-distupgrade python3-distutils python3-gdbm
python3-ldna python3-ldb python3-lib2to3 python3-openvswitch python3-pip
python3-problem-report python3-software-properties python3-tk python3-twisted python3-zipp
python3.10 python3.10-dev python3.10-minimal rfc802 rsync samba-ls
software-properties-common software-properties-gtk ssh sudo systemd systemd-sysv tar tcpdump
tzdata ubuntu-advantage-desktop-daemon ubuntu-drivers-common ubuntu-release-upgrader-core ufw
uno-libs-private unzip update-notifier-common ure util-linux uuid-runtime vim vim-common
vim-runtime vim-tiny wget wpasupplicant xserver-common xserver-xorg-core xxd yaru-theme-gtk
yaru-theme-icon yaru-theme-unity
363 upgraded, 0 newly installed, 0 to remove and 18 not upgraded.
Need to get 817 MB of archives.
After this operation, 91.7 MB disk space will be freed.
Do you want to continue? [Y/n]
```

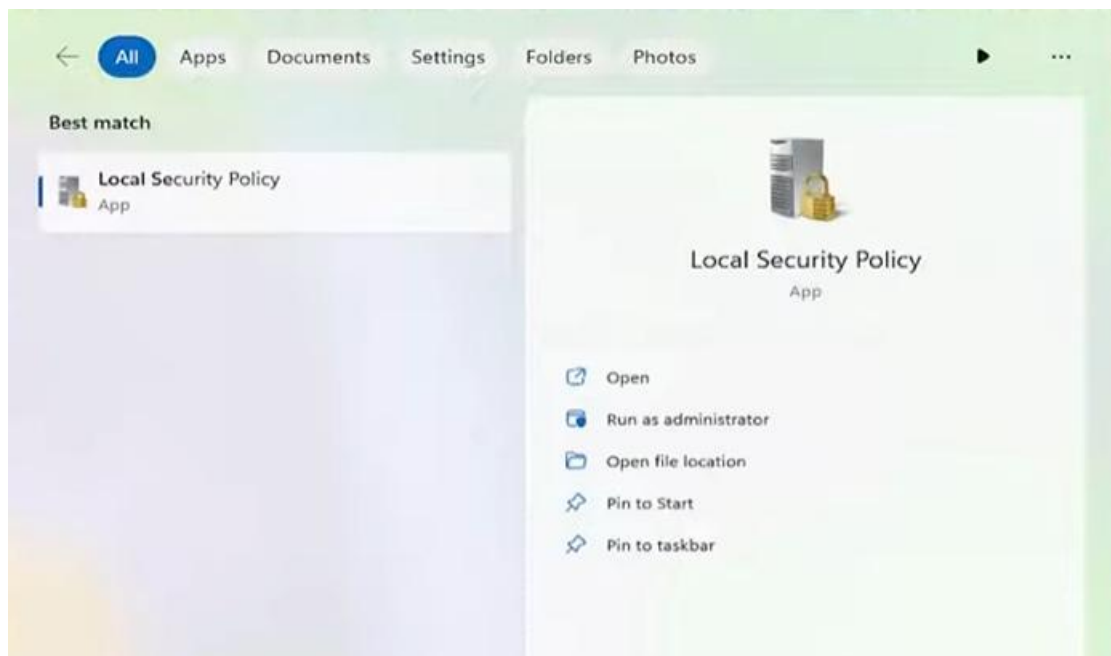
C. Part 2: Windows Local Security Policy

The Local Security Policy in Windows was configured to enforce strong password rules. This included setting password length, complexity, expiration, and history requirements. Account lockout policies were also configured to prevent brute-force attacks by locking accounts after multiple failed login attempts. Audit policies were reviewed to track login activities and system events.

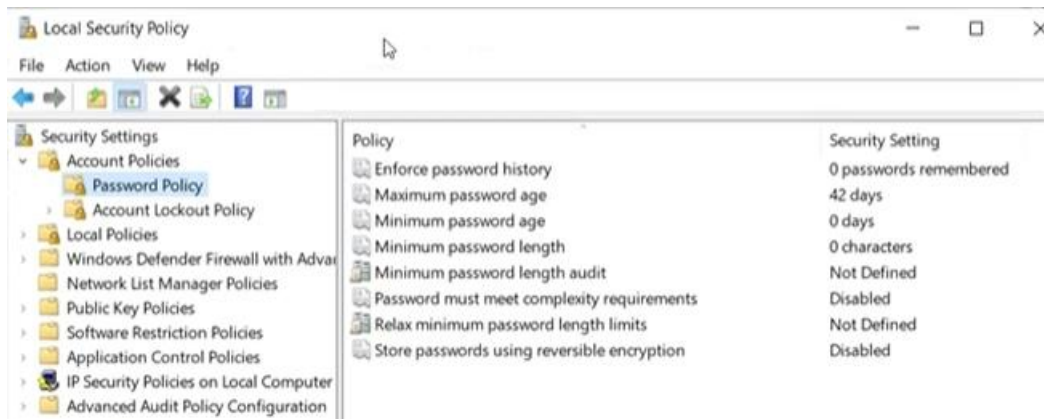
Step 1: Configure Local Password Policy in Windows

You have determined that the security policy for password is as follows:

- A user must use a unique password for at least 2 changes of the password.
 - Passwords must be at least 8 characters.
 - Passwords must be changed every 90 days.
 - Can only change the password once a day.
 - A password must consist of the following four elements:
 - o At least one lower case alpha character.
 - o At least one upper case alpha character.
 - o At least one numerical character.
 - o At least one symbol character.
- a. Navigate to the **Local Security Policy** by first searching for and opening the **Control Panel**.
 - b. Click **Administrative Tools** > Search for **Local Security Policy**.
 - c. Open **Local Security Policy**.



- d. Expand **Account Policies** and click **Password Policy**. Six policies are displayed in the right panel with their associated default security settings.

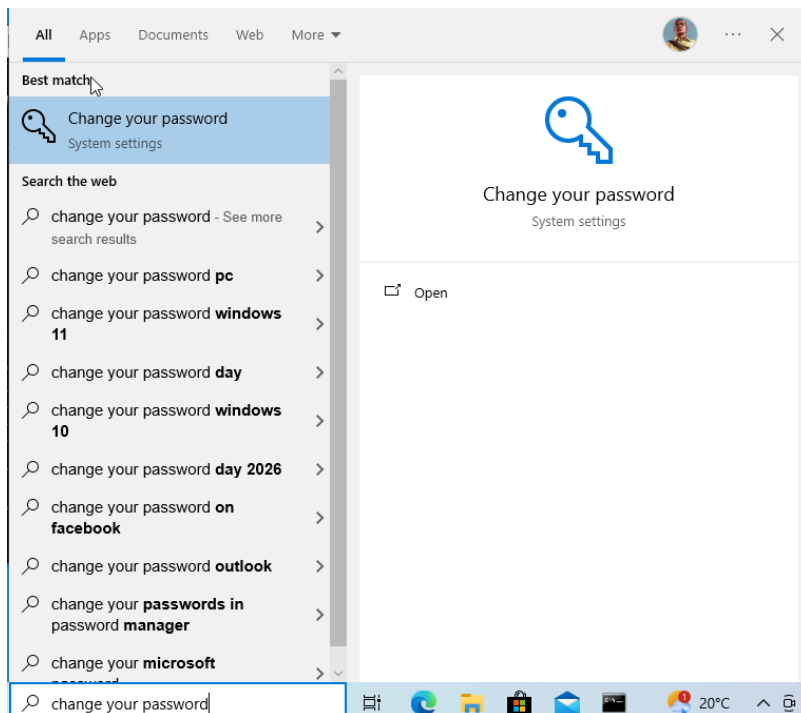


- e. The first policy, **Enforce password history**, is used to set the number of unique passwords the user must enter before being allowed to reuse a password. Double-click **Enforce password history** to open the **Enforce password history Properties** window. Set the value to **2**.
- f. Fill in the values you should set in Local **Security Policy** for the remaining **Password Policy** security settings.

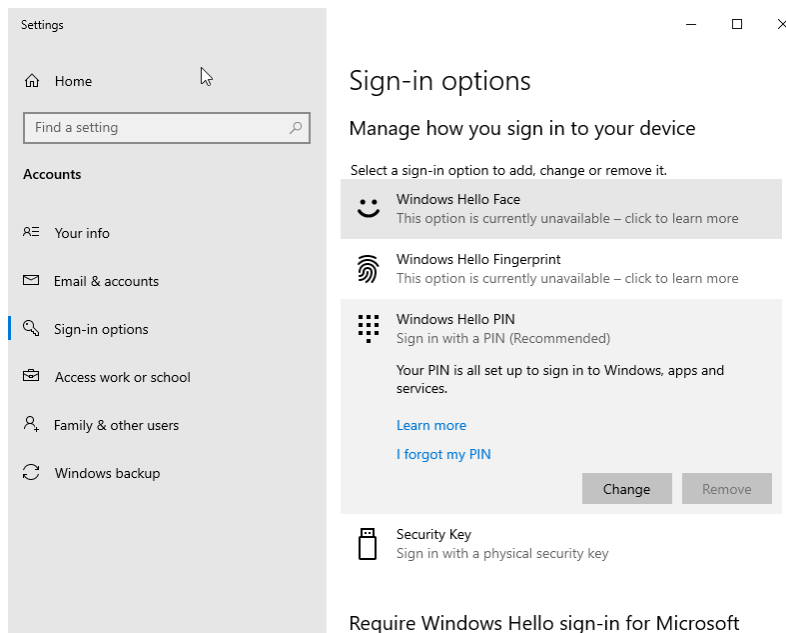
Step 2: Test the password policy security settings.

Test your password policy security settings by attempting to change the password. Try a new password that does not meet the length or complexity requirements.

- a. From the Start Menu, search for **Change your password**.



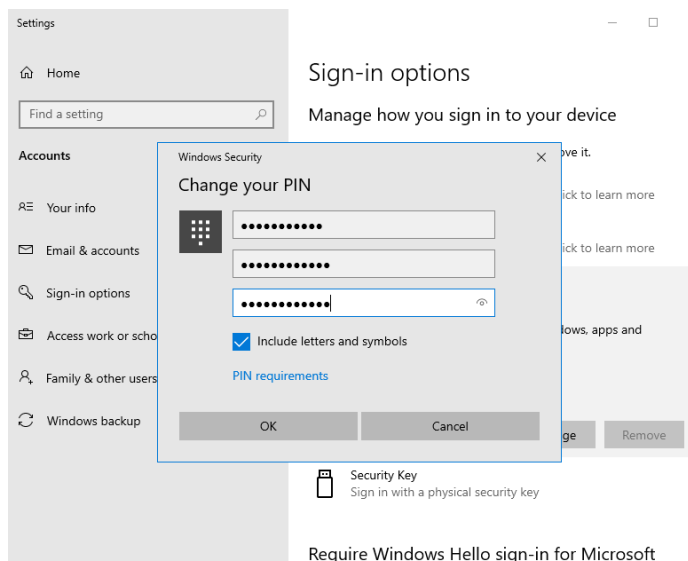
b. Click **Password**. Click **Change**.



c. Enter your current password. Click **Next** to continue.

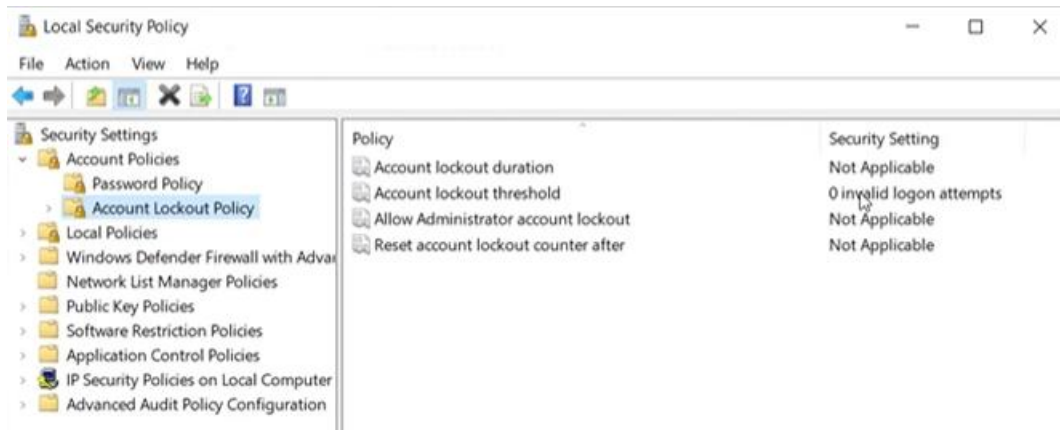
d. Enter your new password twice and provide a password hint. Make sure your new password does not meet the length or complexity requirements that you configured in the previous step. Click **Next** to continue.

e. Click **Finish**. You should be presented with a message that your new password does not meet password policy requirements. Click **Close** to continue.



Step 3: Configure the Account Lockout Policy security settings.

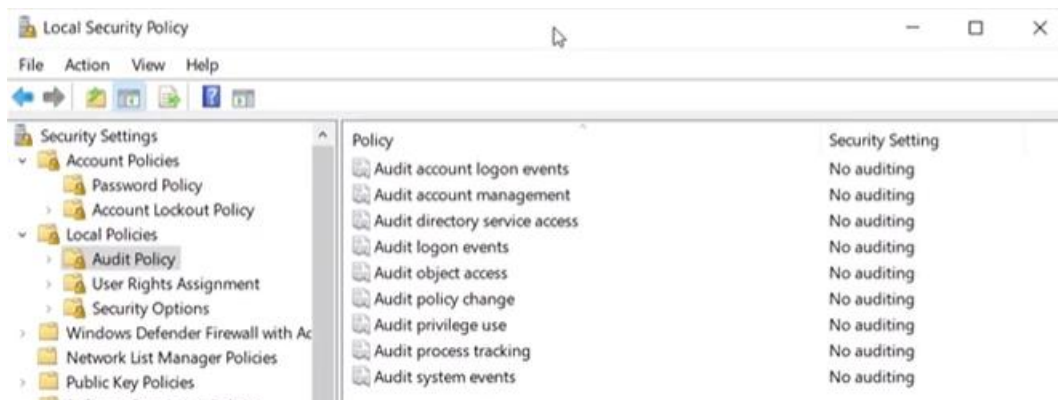
- a. Navigate back to the Local Security Policy window.
- b. In the expanded **Account Policies** and click **Account Lockout Policy**. Three policies are displayed in the right panel with their associated default security settings.



- c. Change the default settings to the following:
 - o A user must wait 10 minutes for the lookout counter to reset.
 - o Users are locked out of the computer after 5 attempts to enter the correct password.

Step 4: Configure the Audit Policy security settings.

- a. Expand the **Local Policies** menu, and then click **Audit Policy**.
- b. Double-click to open the properties window for **Audit account logon events**.
- c. In the **Local Security Setting** tab, notice the check boxes for **Success** and **Failure**.
- d. Click the **Explain** tab to learn about this security setting. Click **OK** to close the **Properties** window.
- e. Continue to review each security setting. Click the **Explain** tab for each and read what it does.

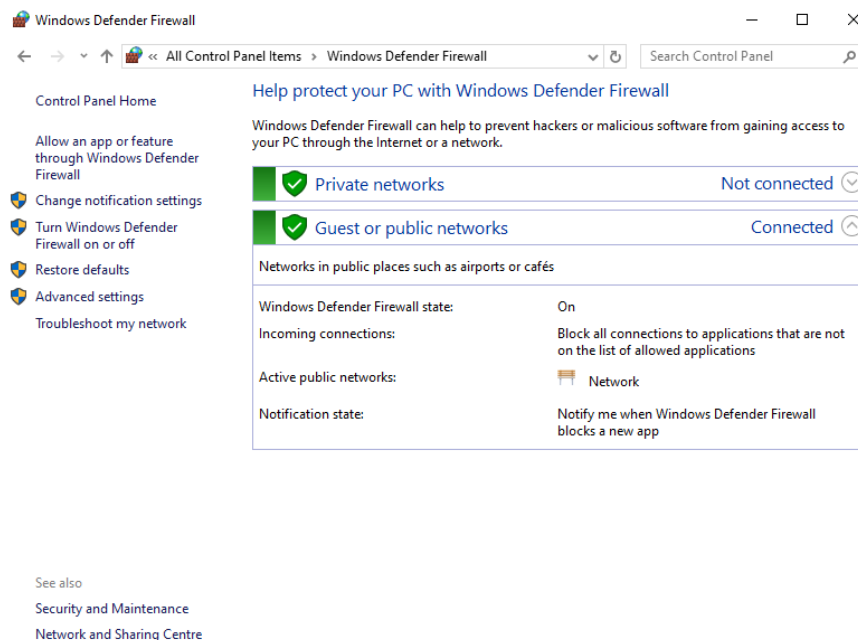


D. Part 3: Firewall Configuration

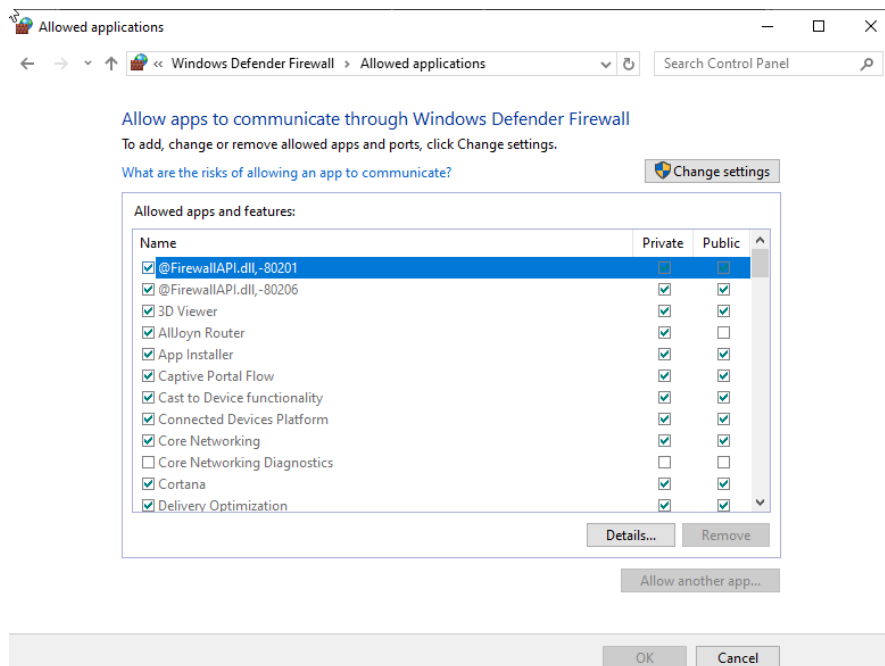
The Windows Defender Firewall was explored and configured to control network traffic. A new inbound rule was created to allow ICMP echo requests (ping). This allowed the Linux machine to successfully ping the Windows host. Firewalls act as security guards by allowing or blocking traffic based on rules.

Step 1: Investigate Windows Defender Firewall

- From the Start Menu, Search for and open **Windows Defender Firewall**. The default state for the Windows Firewall is On.



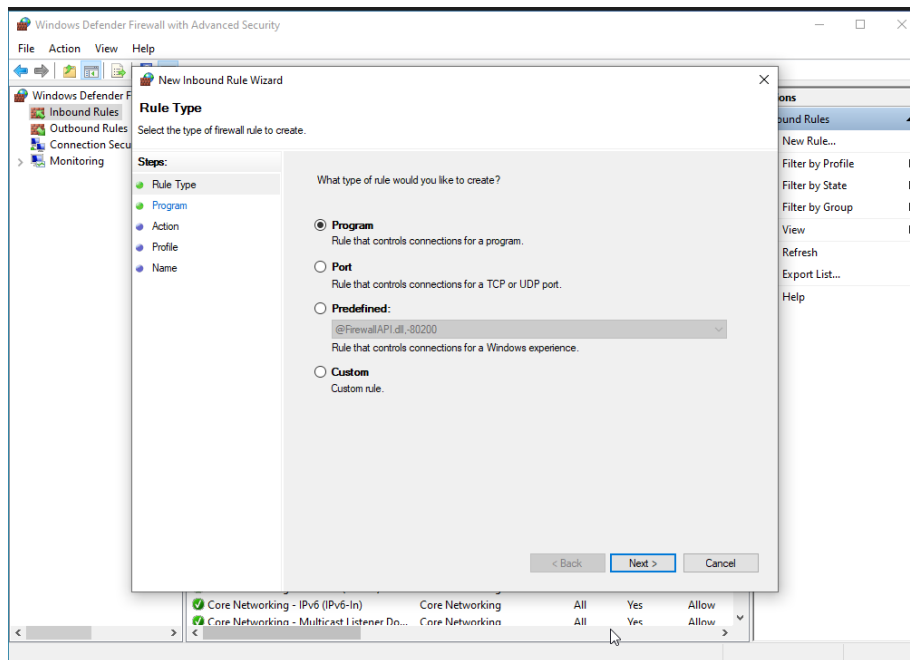
- In the left panel of the window, click **Allow an app or feature through Windows Defender Firewall**. In the **Allowed apps and features** window, programs and services that Windows Firewall is not blocking will be listed with a check mark.



Step 2: Configure the Advanced Security features in Windows Firewall to allow echo requests.

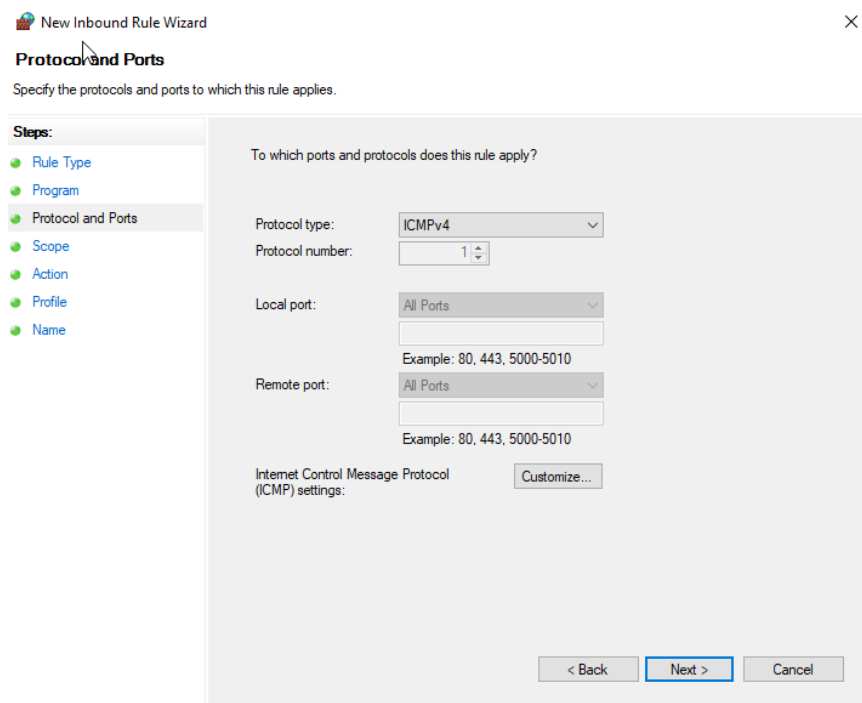
In this step, you will create an inbound rule that will allow echo request packets through the firewall.

- In the left panel of the Windows Defender Firewall, click **Advanced settings**.
- In the **Windows Defender Firewall with Advanced Security on Local Computer**, you can configure Inbound Rules, Outbound Rules, or Connection Security Rules. You can also click **Monitoring** to view the status of configured rules.
- Click **Inbound Rules**, and then in the **Actions** pane, click **New Rule**.

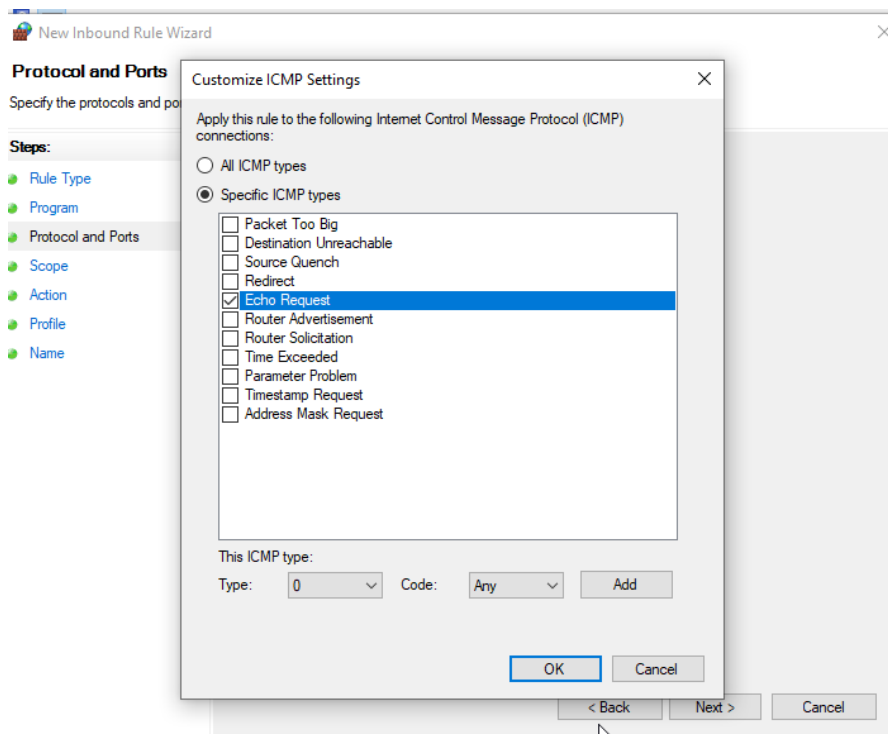


d. In the **New Inbound Rule Wizard**, select **Custom**, and then click **Next** twice. You should now be at the **Protocol and Ports** step.

e. For **Protocol type**, select **ICMPv4**, and then click **Customize**.

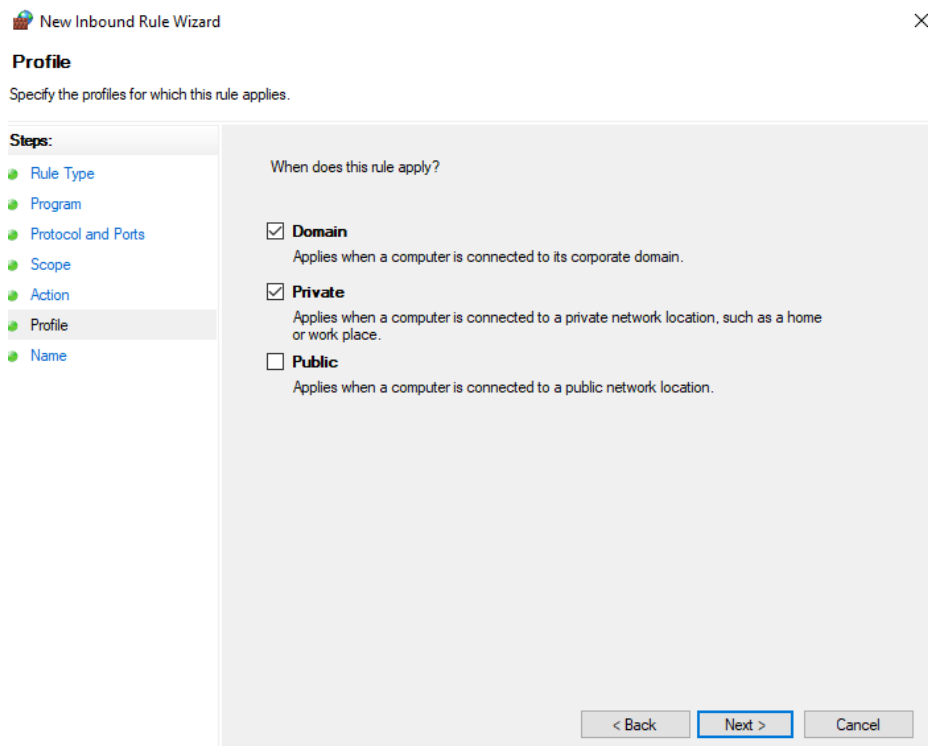


f. In the **Customize ICMP Settings** window, select **Specific ICMP types**, select **Echo Request**, and then Click **OK**.



g. Click **Next** three times. You should now be at the **Profile** step.

h. Deselect **Public** so the Windows PC will not respond to an echo request in a public network location, such as an internet cafe. Click **Next** to continue.



- i. Provide a name for the new **inbound rule** that gives a good description of the rule and click Finish. You should now see **your rule** at the top of the list of Inbound Rules in the **Windows Defender Firewall with Advanced Security** dialog window.
- j. Now the rule has created and enabled. Verify that **CSE-LABVM** can ping the Windows host and receive replies.

```
[analyst@secOps ~]$ ping google.com
PING google.com (142.251.47.110) 56(84) bytes of data.
64 bytes from jnb03s08-in-f14.1e100.net (142.251.47.110): icmp_seq=1 ttl=117 time=40.1 ms
64 bytes from jnb03s08-in-f14.1e100.net (142.251.47.110): icmp_seq=2 ttl=117 time=46.6 ms
```

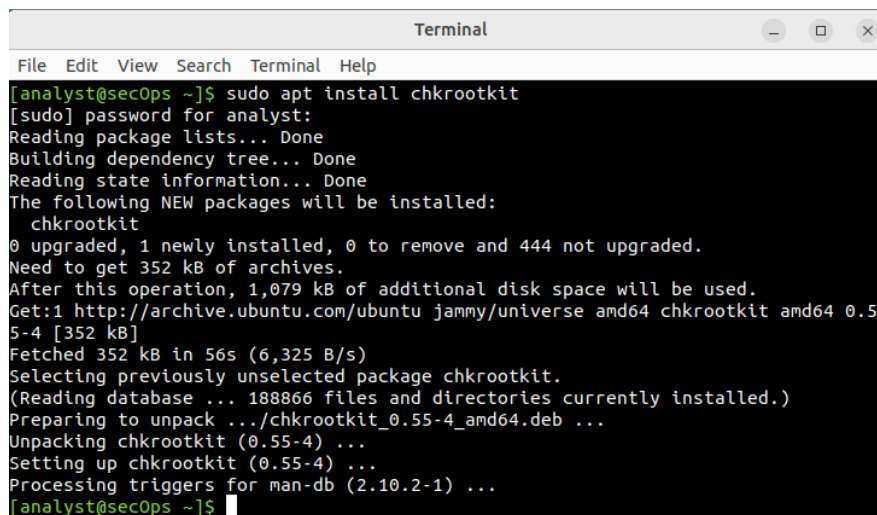
E. Part 4: Installing Security Tools on Linux

Two security tools were installed: chkrootkit and lynis. The command 'sudo apt install chkrootkit' installed a rootkit detection tool. Running 'sudo chkrootkit' scanned the system for hidden malware. The output was filtered using 'grep -i worm' to search for specific threats. Lynis was installed by adding a new repository and then using 'sudo apt install lynis'. Lynis is used for security auditing and system hardening.

Step 1: Install and run chkrootkit

The tool **chkrootkit** is used to check for signs of a rootkit on a local system. Rootkit is a type of malware that can remain hidden on your computer and can be used to cause significant damage to your device by hackers.

- a. In a terminal, enter the command **sudo apt install chkrootkit**. Enter the password **password** when prompted.



```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ sudo apt install chkrootkit
[sudo] password for analyst:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  chkrootkit
0 upgraded, 1 newly installed, 0 to remove and 444 not upgraded.
Need to get 352 kB of archives.
After this operation, 1,079 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 chkrootkit amd64 0.55-4 [352 kB]
Fetched 352 kB in 56s (6,325 B/s)
Selecting previously unselected package chkrootkit.
(Reading database ... 188866 files and directories currently installed.)
Preparing to unpack .../chkrootkit_0.55-4_amd64.deb ...
Unpacking chkrootkit (0.55-4) ...
Setting up chkrootkit (0.55-4) ...
Processing triggers for man-db (2.10.2-1) ...
[analyst@secOps ~]$
```


- b. Enter the command **sudo chkrootkit** to run a rootkit check.

```
[analyst@secOps ~]$ sudo chkrootkit
ROOTDIR is '/'
Checking `amd'... not found
Checking `basename'... not infected
Checking `biff'... not found
Checking `chfn'... not infected
Checking `chsh'... not infected
Checking `cron'... not infected
Checking `crontab'... not infected
Checking `date'... not infected
Checking `du'... not infected
Checking `dirname'... not infected
Checking `echo'... not infected
Checking `egrep'... not infected
Checking `env'... not infected
Checking `find'... not infected
Checking `fingerd'... not found
Checking `gpm'... not found
Checking `grep'... not infected
Checking `hdparm'... not infected
Checking `su'... not infected
Checking `ifconfig'... not infected
Checking `inetd'... not infected
Checking `inetdconf'... not infected
```

- c. The output can be filtered to look for interested strings, such as worm. The command **chkrootkit** can be piped together with the command **grep** with the option **-i** to ignore the case distinction in the strings of interest.

```
[analyst@secOps ~]$ sudo chkrootkit | grep -i worm
Searching for LPD Worm files and dirs... nothing found
Searching for Ramen Worm files and dirs... nothing found
Searching for Adore Worm... nothing found
Searching for ShitC Worm... nothing found
Searching for Omega Worm... nothing found
Searching for Sadmind/IIS Worm... nothing found
Searching for TC2 Worm default files and dirs... nothing found
! analyst 3037 pts/0 grep -i worm
[analyst@secOps ~]$
```

Step 2: Install Lynis

lynis is security tool for systems running Unix-based OS, such as Linux and macOS. **lynis** will be used later in another activity to harden a Linux system. The application **Lynis** is maintained by [CISOfy](#). In this step, we will add the software repository and install Lynis.

- a. Copy and paste the following command into a terminal to import the key from the CISOfy keyserver. This key is required to verify the integrity of your download when you download **lynis**:

```
[analyst@secOps ~]$ sudo apt-key adv --keyserver keyserver.ubuntu.com --recv-key
s 013baa07180c50a7101097ef9de922f1c2fde6c4
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (s
ee apt-key(8)).
Executing: /tmp/apt-key-gpghome.4ZyAcHH4l0/gpg.1.sh --keyserver keyserver.ubuntu
.com --recv-keys 013baa07180c50a7101097ef9de922f1c2fde6c4
gpg: keyserver receive failed: No data
[analyst@secOps ~]$
```

b. Copy and paste the following command into a terminal to add the **lynis** repository maintained by CISOfy.

```
[analyst@secOps ~]$ echo 'deb https://packages.cisofy.com/community/lynis/deb/ s
table main' | sudo tee /etc/apt/sources.list.d/cisofy-lynis.list
deb https://packages.cisofy.com/community/lynis/deb/ stable main
[analyst@secOps ~]$
```

c. Perform an update after adding a new repository. At the prompt, **enter sudo apt-get update**.

d. Use the command **apt install lynis** to install Lynis.

```
[analyst@secOps ~]$ sudo apt install lynis
[sudo] password for analyst:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-l10n gksu
  | kde-runtime | ktsuss
The following NEW packages will be installed:
  lynis menu
0 upgraded, 2 newly installed, 0 to remove and 499 not upgraded.
Need to get 581 kB of archives.
After this operation, 3,164 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

e. To verify the installed version, enter the command **lynis show version** at the terminal.

```
[analyst@secOps ~]$ lynis show version 3.0.6
3.0.7
```

```
[analyst@secOps ~]$ sudo apt-cache policy lynis
lynis:
  Installed: 3.0.7-1
  Candidate: 3.0.7-1
  Version table:
  *** 3.0.7-1 500
       500 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages
       100 /var/lib/dpkg/status
[analyst@secOps ~]$
```

F. Conclusion

This lab demonstrated how to secure systems by keeping them updated, enforcing strong policies, managing firewall rules, and using security tools. These practices are essential for protecting systems from cyber threats.